

Eidgenössische Technische Hochschule Zürich
Ecole polytechnique fédérale de Zurich
Politecnico federale svizzero di Zurigo
Swiss Federal Institute of Technology Zurich
Pysical Chemistry Laboratory
8092 Zürich, Switzerland
http://www.esr.ethz.ch

Multi-photon Resonances in Pulse EPR

I. Gromov and A. Schweiger

41st Rocky Mountain Conference on Analytical Chemistry,
Denver 1999

Introduction

The interaction of electromagnetic radiation with matter, where several photons are simultaneously absorbed or emitted, is the physical basis of many effects, as multi-photon absorption, emission and ionization, Raman scattering, second harmonic generation, etc. [1, 2]. Multi-photon processes are well studied and widely used in coherent optics. At microwave and radio frequencies these processes are less known and consequently less used. The use of multi-photon resonances in two-level systems involving an odd number of photons has been demonstrated in cw EPR [3] and pulse NMR [4] spectroscopy. Two-photon resonance was observed in cw EPR [5] and has been used to monitor transient nutations in an electron spin system [6]. In the present work, for the first time odd and even multi-photon resonances observed in pulse EPR, are described. It is also shown that Floquet theory [7], usually applied for the description of effects where an odd number of photons is involved, can be extended to an even number of photons. Possible applications of the multi-photon transitions in paramagnetic systems are proposed.

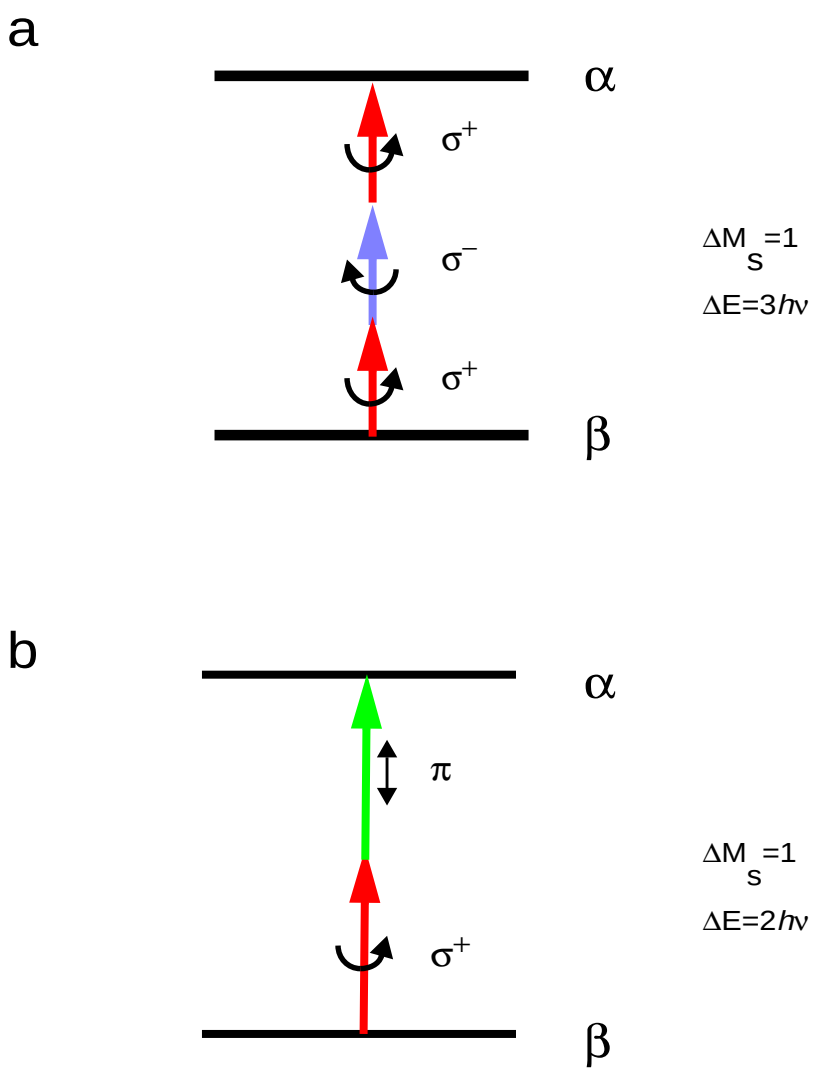
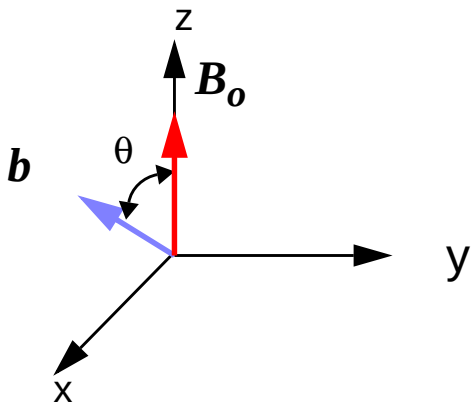


Fig.1. Multi-photon transitions between the two states of an S=1/2 spin system with energies E_α, E_β. Right and left circularly polarized radiation, σ^+ and σ^-, induces three-photon transitions (a); linear polarized (with respect to B_0) π radiation is necessary for two-photon processes (b).

Theoretical background

Conside a spin system, resonant at $\omega_0 = \gamma B_0$, which is driven by a microwave field b , oscillating at angular frequency $\omega < \omega_0$.



The Hamiltonian of the system is given by

$$H_c(t) = \begin{bmatrix} E_\alpha + b_z & b_r \\ b_r & E_\beta - b_z \end{bmatrix}, \qquad (1)$$

where b_z and b_r are the oscillating field || and $\perp$ $\boldsymbol{B_0}$.

Following Shirley [7], we expand (1) in Fourier series and construct the *Floquet Hamiltonian* H_F :

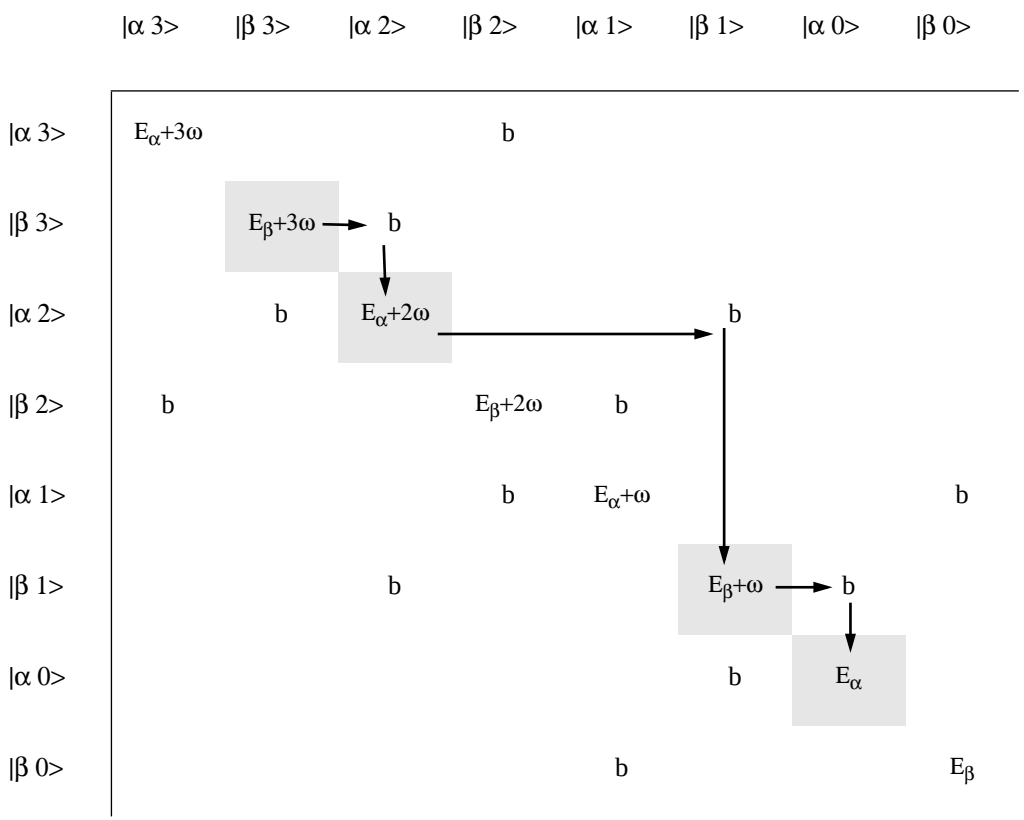
$$\langle \alpha n | H_F | \beta m \rangle = H_{\alpha\beta}^{n-m} + n\omega\delta_{\alpha\beta}\delta_{nm}, \qquad (2)$$

where α , β represent the spin states and *n,m* - the Fourier components.

Analysing a behaviour of the H_F eigenvalues yields the probabilities of the multi-photon transition.

Map of the multiple transition between energy levels of the Floquet Hamiltonian

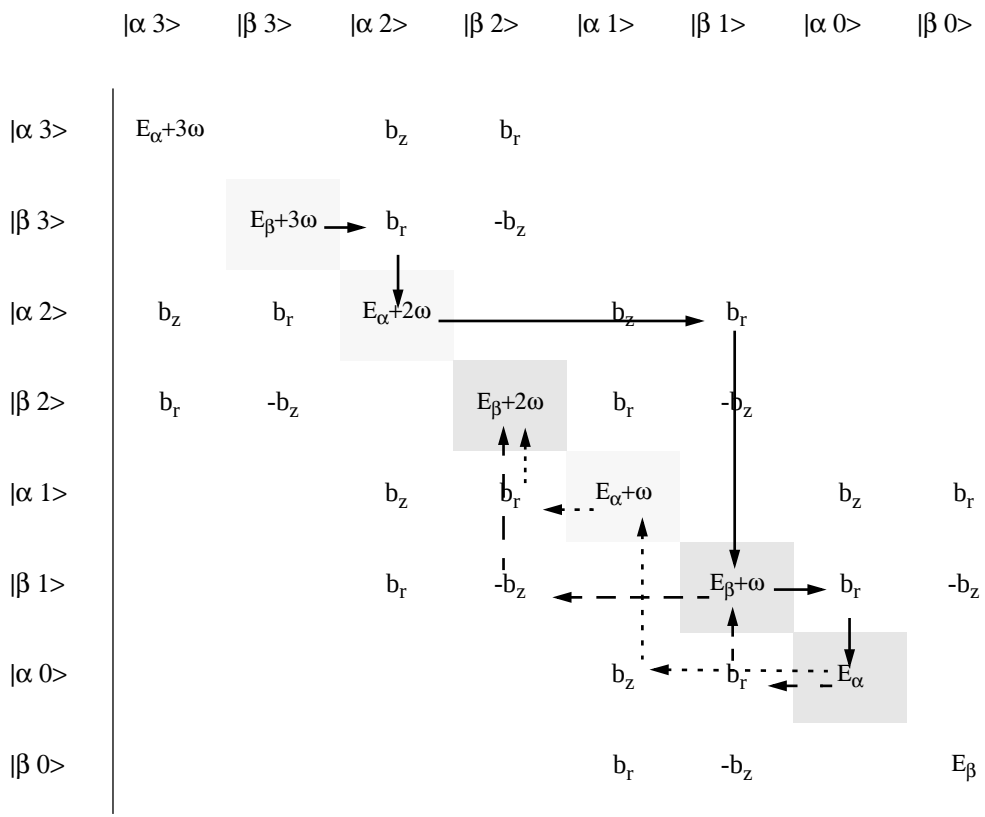
$b \perp B_0$ Odd resonance



→ 3-photon transition pathway;

3-photon effective field $\omega_{eff} \sim \frac{b^3}{B_0^2}$, for $b \ll \omega$, $E_\alpha \approx E_\beta + n\omega$

$\theta = \angle(b, B_0) = 45^0$ Even resonances



Pathways: ·····▶ – – –▶ 2-photon transition; ——▶ 3-photon transition;

2-photon transition moment: $\omega_{eff} \sim \frac{b^2}{B_0}$, for $b \ll \omega$, $E_\alpha \approx E_\beta + n\omega$

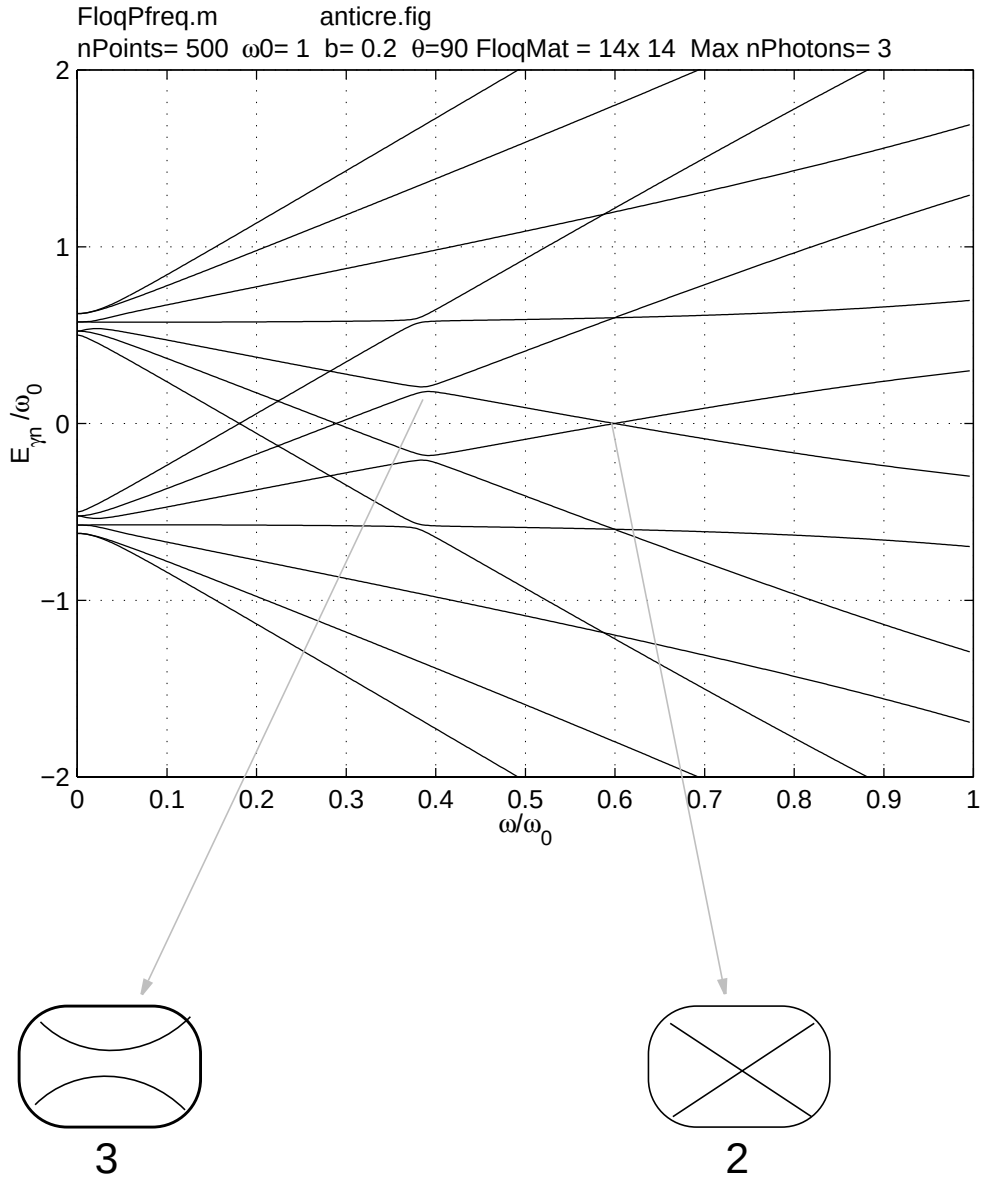


Fig. 2. Eigenvalues of the Floquet Hamiltonian (2) for $b \perp B_0$, as a function of ω with: Eα,β= ± ω0 / 2; ω=1; b=0.5. The diagram shows anti-crossings near $\omega 0/\omega = 3$ which are responsible for 3-photon resonances. Even anticrossings are not observed.

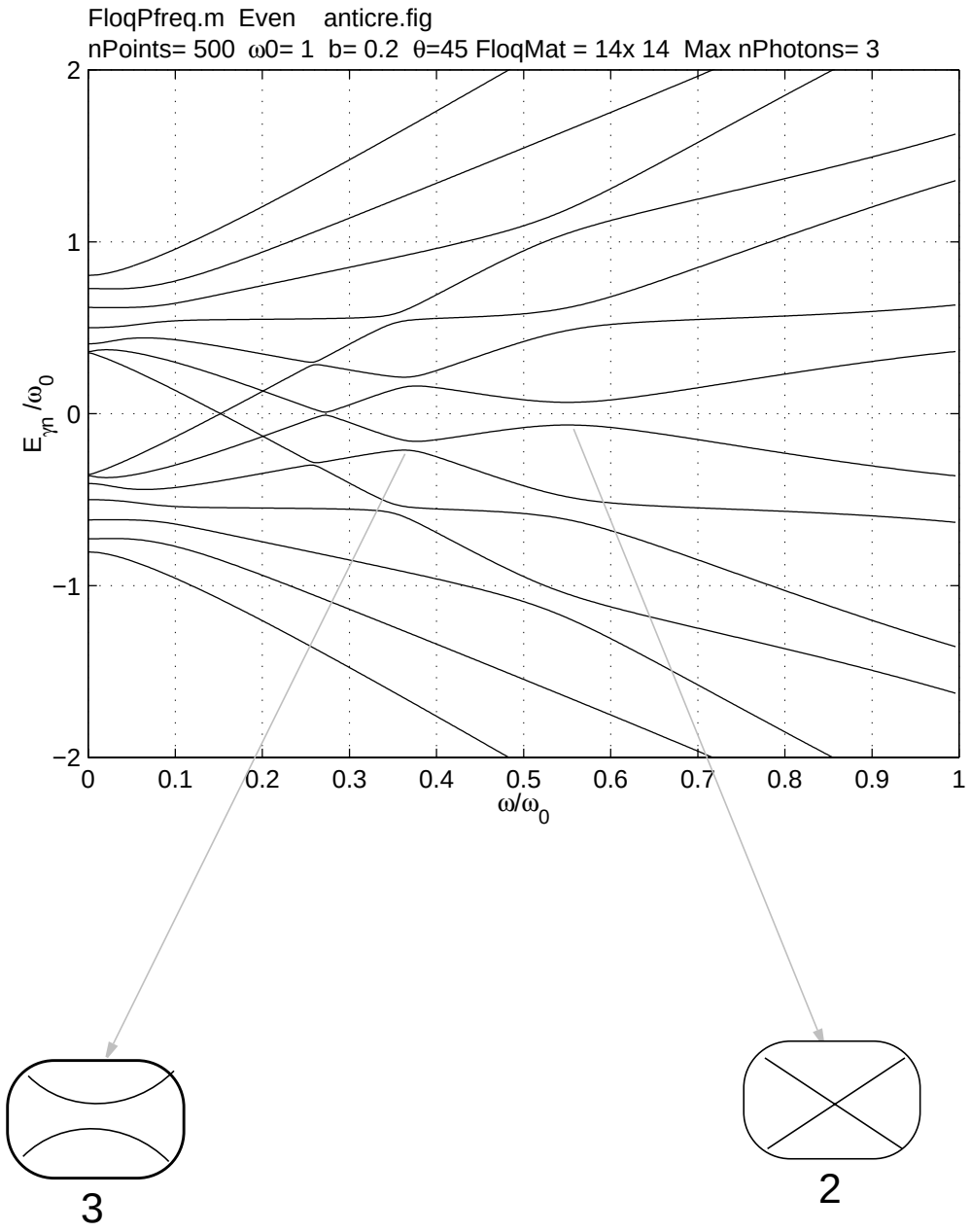
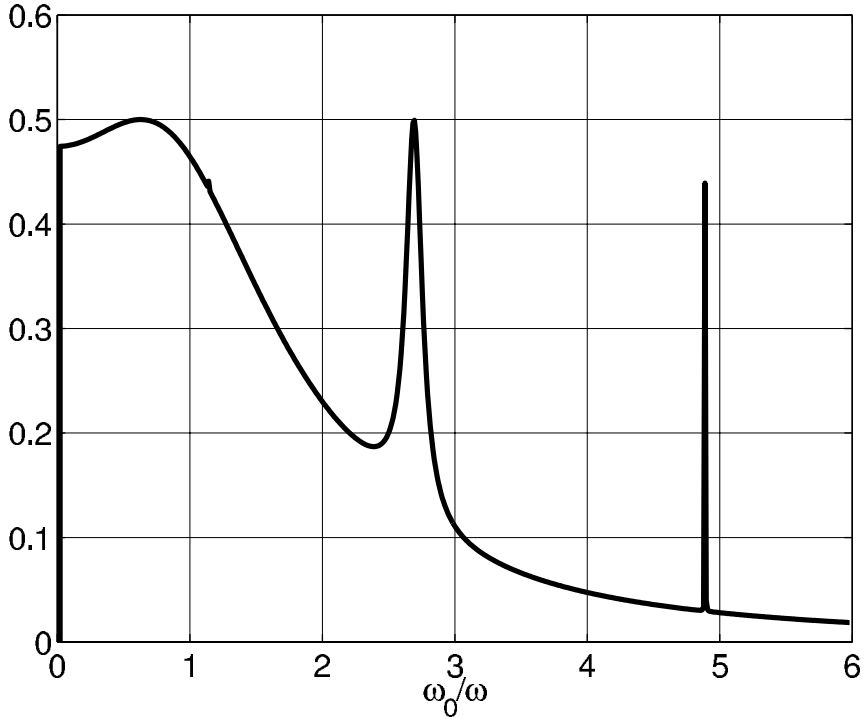


Fig. 3. Eigenvalues of the Floquet Hamiltonian (2) for $\theta = \angle(b, B_0) = 45^0$, as a function of ω with: $E_{\alpha,\beta} = \pm \omega_0 / 2$; ω=1; b=0.5. The diagram shows odd and even anticrossings.

a



b

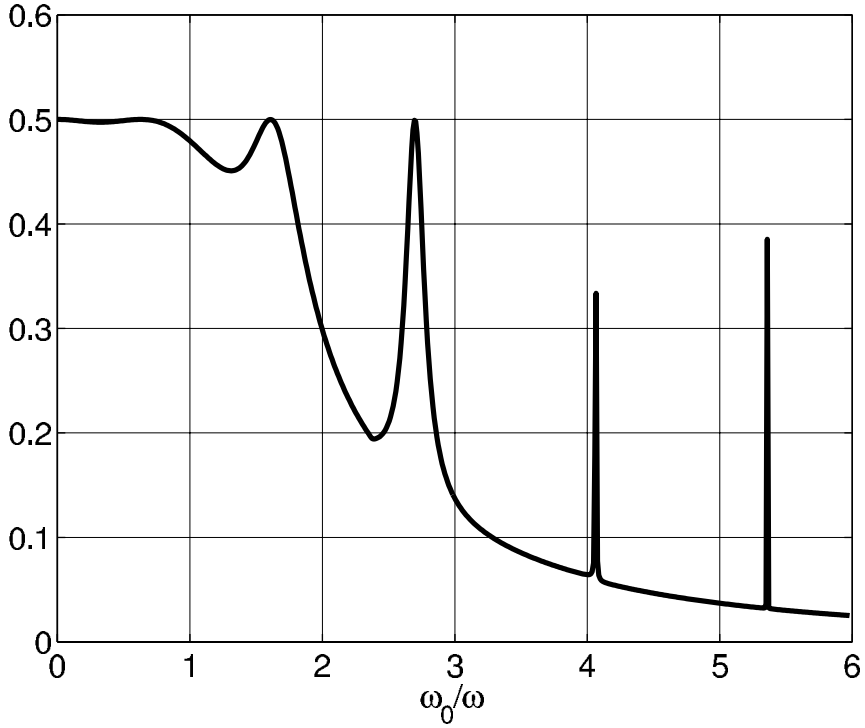
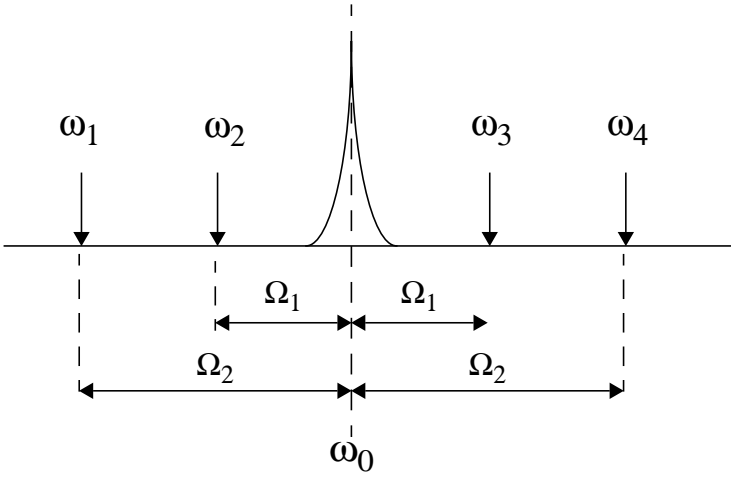


Fig.4. Transition probabilities between states of the Floquet Hamiltonian for $b \perp B_0$; ω=1; b=0.5 (**a**), and $\angle(b, B_0) = 45^0$; ω=1; b=0.75 (**b**), calculated by using the first derivatives of the energy levels as a function of ω_0.

Experimental results

Odd resonances

Tetrachromatic excitation

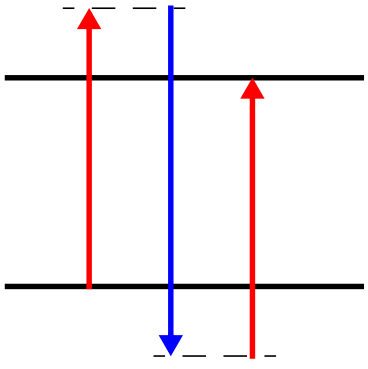


Resonance frequencies: $\omega_{res} = \omega_0 + n_1\Omega_1 + n_2\Omega_2$

$$n_1 = n + 2m + 1; \quad n_2 = \mp n$$

3-photon transition diagram:

$[n_1, n_2] = [2, -1]$



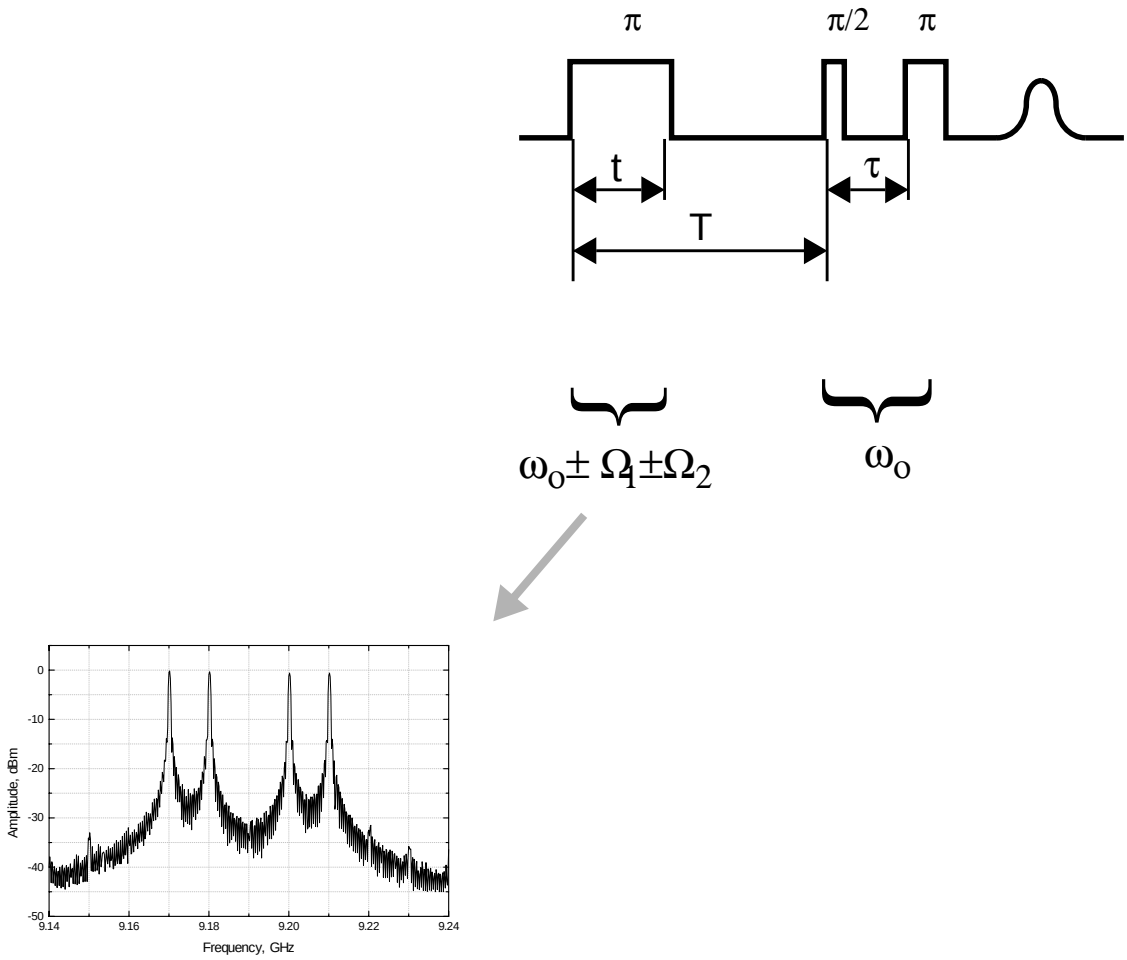


Fig. 5. Inversion-recovery pulse sequence. The multi-photon resonances, induced by a tetrachromatic inversion pulse, are detected using a single-photon two-pulse echo. Insert shows the spectrum of the tetrachromatic pulse measured with a HP8565 analyzer. The instrumental intermodulation distortion is less than -35 dBc.

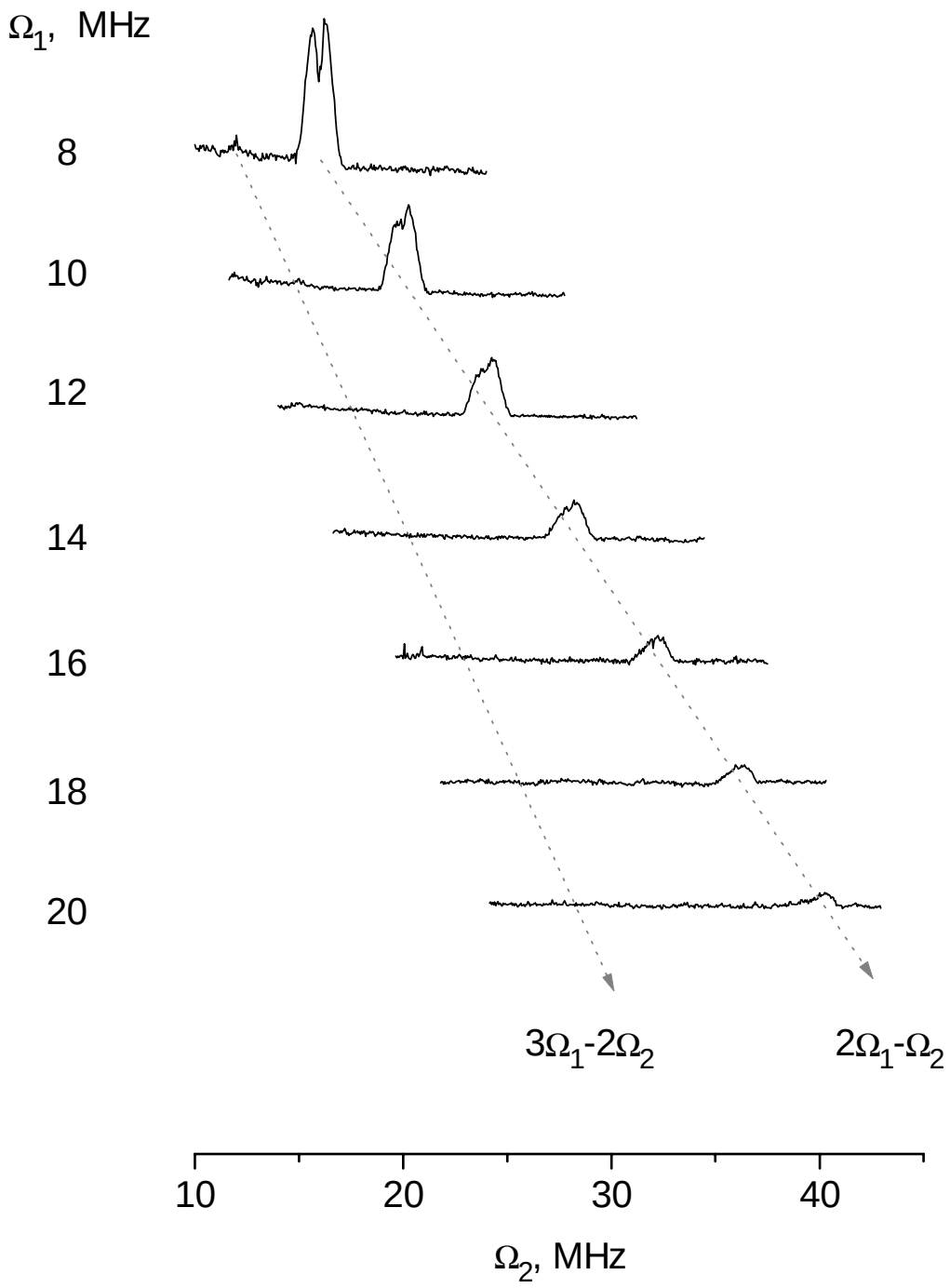
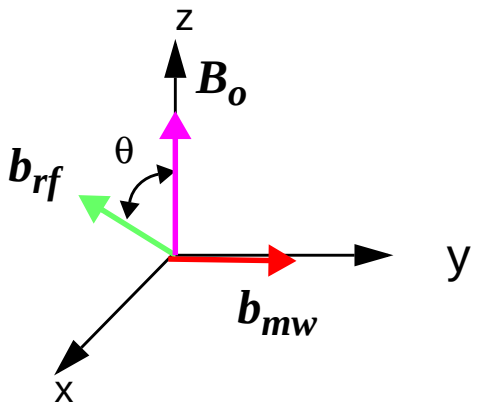


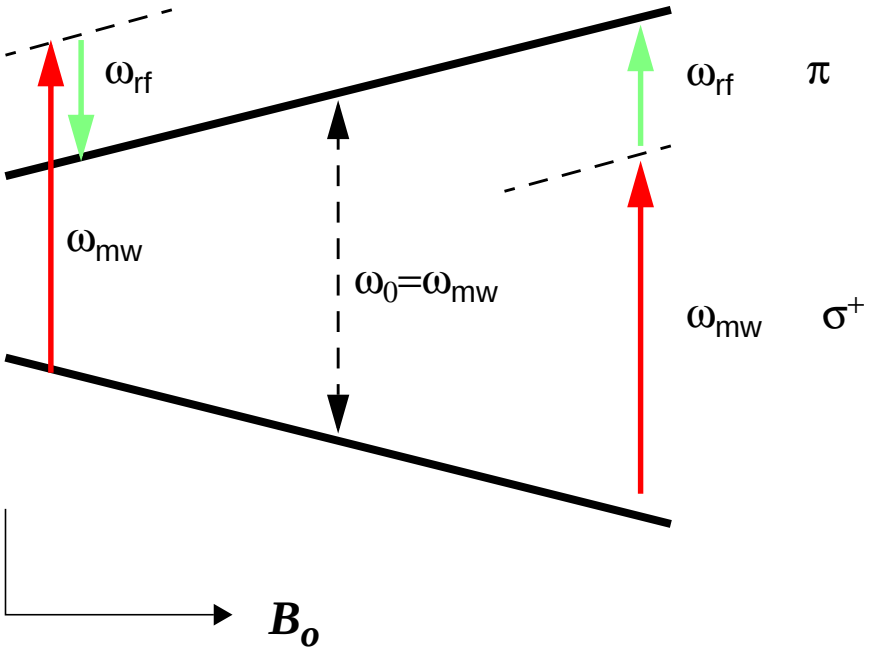
Fig. 6. Integrated echo intensity of γ-irr. quartz glass (Herasil) as a function of Ω_2 at different fixed Ω_1 values of the tetrachromatic pulse. Experimental conditions: t=2 μs; T=11μs; τ=1.5 μs; ν_{mw}=9.035 GHz; B_0=322.6 mT. 3-photon resonances are observed when condition $2\Omega_1$-Ω_2 is fulfilled. Weak 5-photon resonances $3\Omega_1$-$2\Omega_2$ are visible at the two top traces.

Even resonances

RF_z + MW_y pulse excitation



2-photon transition diagram:



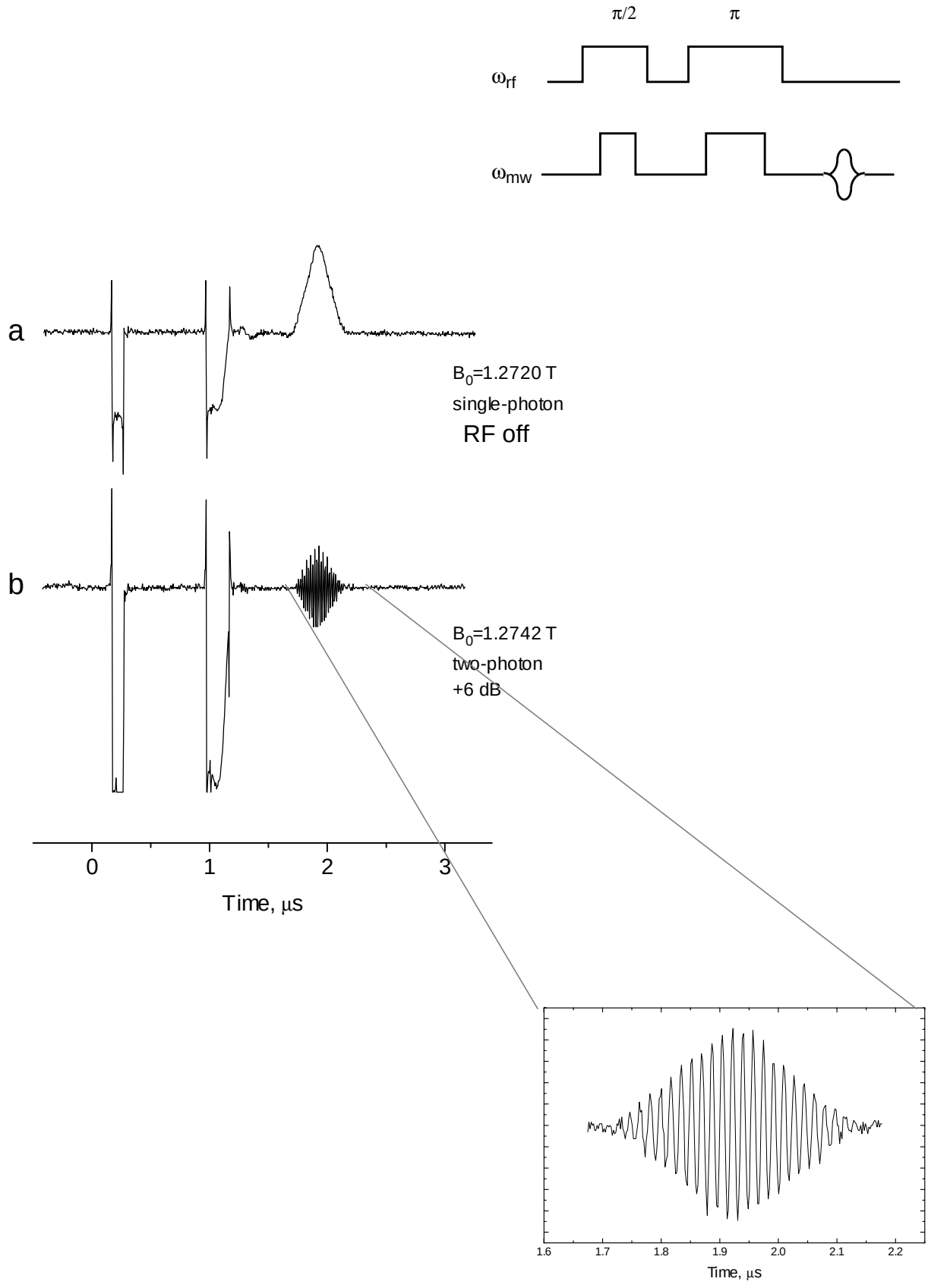


Fig. 7. Two-photon echo of coal (Bruker test sample) at Q-band measured at room temperature. Experimental conditions: $b_{rf} \parallel B_0$; ν_{rf}=60 MHz (2.14 mT); ν_{mw}=35.35 GHz; b_{rf}=0.5 mT; b_{mw} =0.3 mT for two-photon echo.

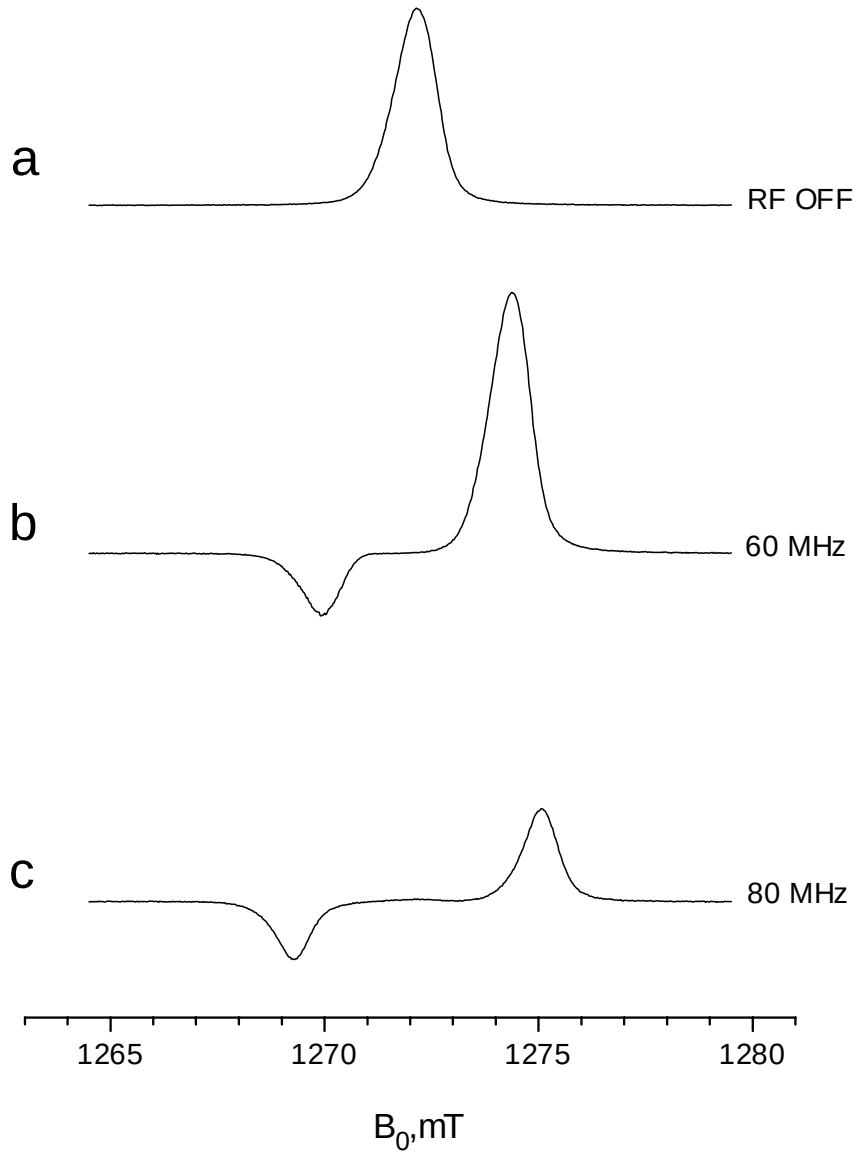


Fig. 8. Echo-detected single- and two- photon spectra of coal. (a) single-photon spectrum. (b,c) Two-photon spectra with ν_{rf}=60 MHz (b), and 80 MHz (c). Experimental conditions see Fig.7.

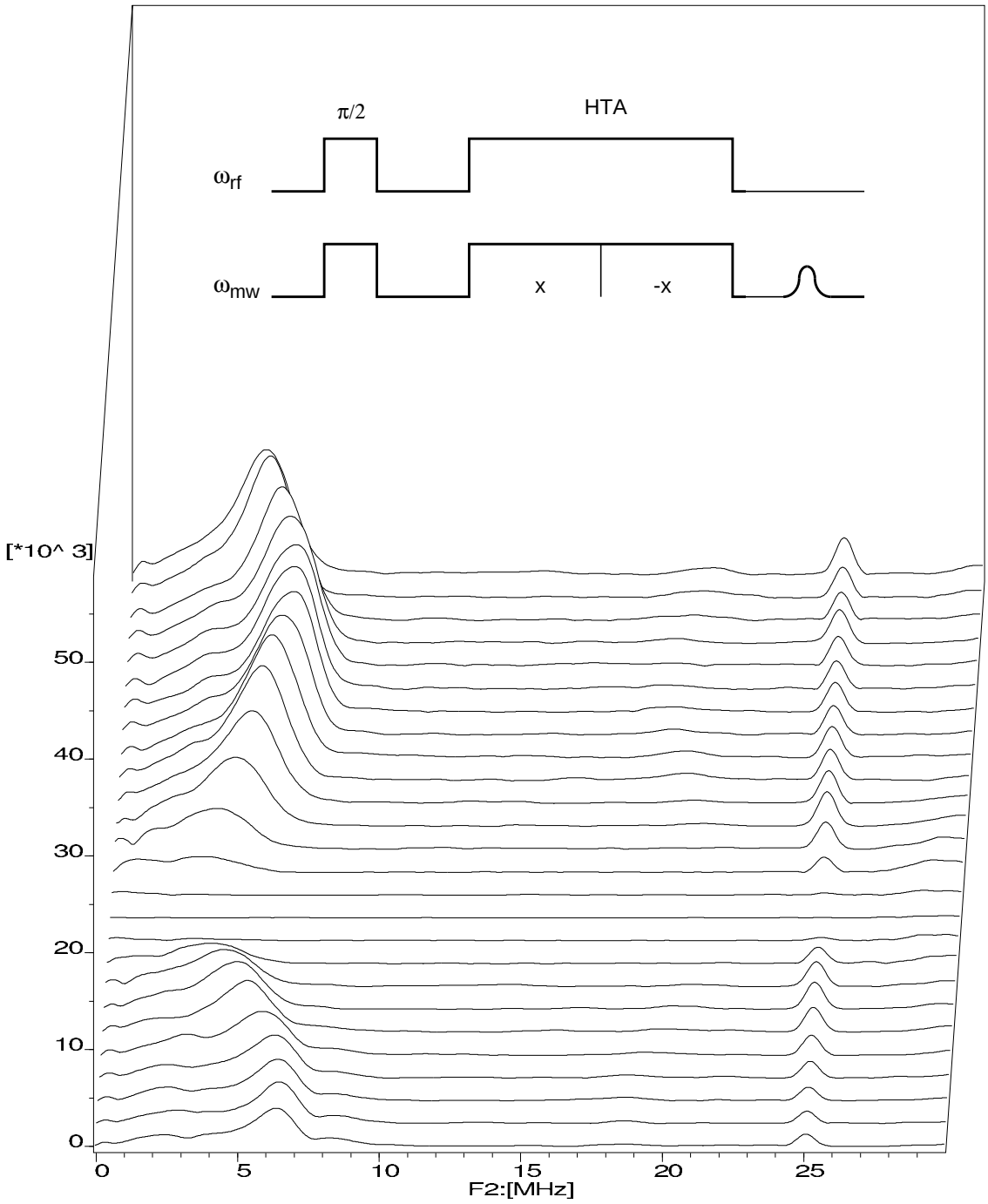


Fig. 9. Two-photon nutation spectra of γ-irr. quartz glass (Herasil) at different orientations of the radio frequency field relatively to the static magnetic field. Experimental conditions: ν_{rf}=25 MHz (0.93 mT); ν_{mw}=9.62 GHz; $b_{rf}=b_{HTA}$=0.4 mT; B_0=344.46 mT; length of the HTA pulse is 5.12 μs.

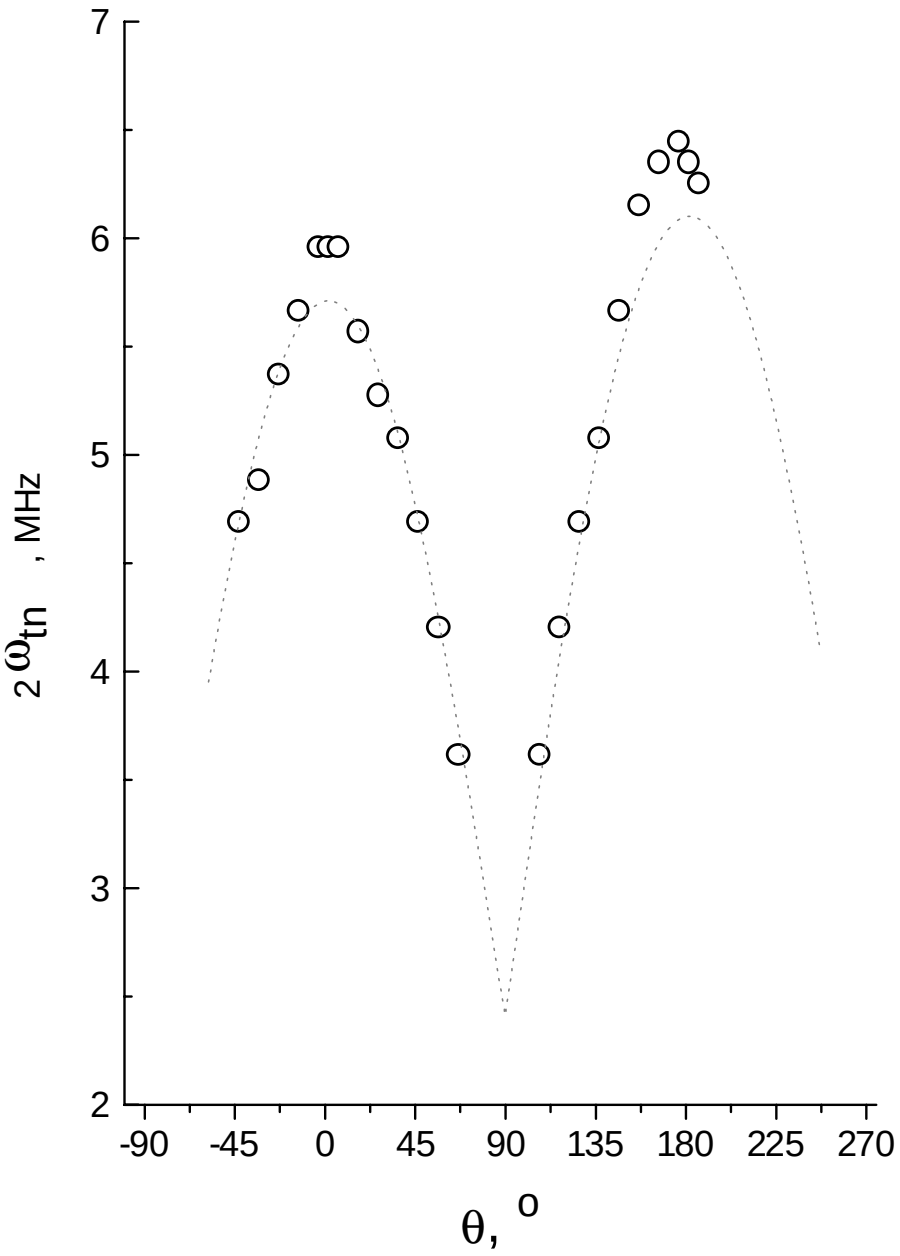


Fig. 10. Nutation frequency as a function of the angle between radio frequency field and static magnetic field. The frequency of two-photon nutation reaches a maximum when fields are parallel.

Discussion

Performed experiments demonstrate that multi-photon signals can easily be detected in pulse EPR and consequently multi-photon method can be used in actual practice, for example: in high field spectroscopy without high frequency, excitation of the multiquantum transition, etc.

Acknowledgments

I.G. gratefully acknowledge technical assistance by Jörg Forrer.

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